

## A New Approach in Cooperative Decision Making in Multi-Agent Systems Inspired by Human Visual Cortex

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### Abstract

In this paper, a new approach in decision making process inspired by human visual cortex has been proposed. In this approach knowledge of a group of agents (training data) will be used for decision-making. The proposed approach tries to meet two fundamental features, i.e., robustness and specificity. The hierarchical model that has been represented in this work, tries to extract the knowledge about the behavior of the system from the training data set by finding the similar training data points. In this model the behavior of the system is governed by the clusters of training data points that in fact every cluster act as an expert. For every new data point, these experts try to predict the label of the corresponding data point and the result of the system is the aggregation of the predictions of different experts. This hierarchical model has been designed inspired by a computational model of object recognition in human cortex. The approach has been used to forecast Mackey-Glass time series and has shown acceptable results.

### 1. Introduction

Human brain is able to recognize the objects independent of the size, rotation and distance. The cognitive task of the human vision system has two fundamental features which are specificity and invariance. Riesenhuber et al [1], has introduced a hierarchical approach for object recognition inspired by the hierarchical nature of the human visual cortex. Concerning the fundamental features of human visual cortex, they have introduced a new set of features which exhibits better trade-off between invariance and selectivity in compare to other approaches in object-recognition. Inspired by Riesenhuber's work and human visual cortex we have proposed an approach in cooperative decision making in a multi-agent system. The presented model, is a hierarchical model which follows the standard model of object recognition in primate cortex [1,2] that assumes the first few hundred milliseconds of visual processing in primate cortex follows a feed forward hierarchy and in each stage the receptive field of neurons get larger. In the proposed model, we try to follow this hierarchical model to generalize the features and make knowledge based clusters which govern the behavior of the system. Every cluster in fact encapsulates partial knowledge of the system. This approach could be very appropriate in cooperative decision making in multi-agent systems in which each agent access part of the knowledge. In this work, through training phase, the experiences of the model is made. These experiences are grouped as some clusters which act like experts to forecast the best reaction for a new action that enters to the system. Each expert region is characterized by the knowledge that has been stored in it (indicator knowledge). For every new action (data), all the experts offer their decisions and then by combining these decisions the final decision will be made.

A reliable decision making model is a model that is able to forecast with minimal error to yield good preparation for the future and serves as a good decision-making. So the proposed model will be used to predict time series in which the last values of the series will be used as the partial knowledge of the agents (training data). The model has been used to predict Mackey-Glass time series. Mackey-Glass equation is a non-linear delay differential equation, one form of which is as below:

$$\frac{ds(t)}{dt} = \frac{0.2s(t-\tau)}{1+s^{10}(t-\tau)} - 0.1s(t) \quad (1)$$

Where it shows chaotic behavior when  $\tau > 17$ . The behavior of this equation has been shown in figure 1, 2 for different amounts of  $\tau$ .

### 2. The Quantitative Model of Human Visual System

Human brain is one of the best object recognition systems and designing a system, which is able to recognize patterns as well as human brain, is a very temptation idea. There are a lot of studies about human vision system as one of the best vision machine in the last years [1-6]. The recently works in machine vision design have been concentrated on designing of vision systems which are inspired by human visual cortex. The number of references and the amount of information about human vision system motivated us to propose a cooperative decision making approach inspired by human vision system. In the designed decision maker system, we follow the standard model of object recognition in primate cortex [1], which in fact is a quantitative model of human visual system. This quantitative model which has been designed concerning the general organization of visual cortex and considering this assumption that the system must meet two fundamental requirements, i.e. specificity and noise tolerance has been accepted by most visual neuroscientists.

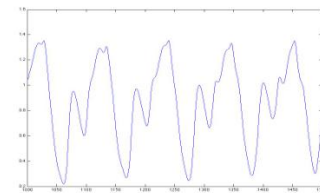


Figure 1 Mackey-Glass time series with  $\tau = 1$

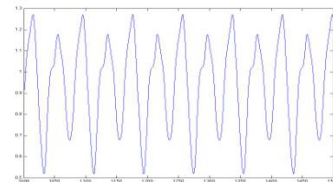


Figure 2 Mackey-Glass time series with  $\tau = 30$

In this model it has been stated that first few hundreds milliseconds of visual processing in primate cortex follows a mostly feed forward hierarchy. At each stage, the receptive fields of neurons (i.e., the part of the visual field that could potentially elicit a neuron's response) tend to get larger along with the complexity of their optimal stimuli (i.e., the set of stimuli that elicit a neuron's response) [2]. This model consists of four layers of computational units where *simple* S units, which combine their inputs with Gaussian-like tuning to increase object selectivity, alternate with *complex* C units, which pool their inputs through a maximum operation, thereby introducing gradual invariance to scale and translation (S and C units are depicted in figure 3.). The model has

been able to quantitatively duplicate the generalization properties exhibited by neurons in inferotemporal monkey cortex (the so-called view-tuned units) that remain highly selective for particular objects (a face, a hand, a toilet brush) while being invariant to ranges of scales and positions [2]. By improving the stages of the hierarchy, the size of the receptive field as well as the complexity of the optimal stimuli for the neurons, will increase. The model originally used a very simple static dictionary of features (for the recognition of segmented objects) although it was suggested in [1] that features in intermediate layers should instead be learned from visual experience.

Expanding the standard model of object recognition in primate cortex, Serre et al. [2] has proposed to design a 4-layer hierarchical model, in which each layer operates as below:

- S1 layer is the first layer and has a direct access to the images. The output of this layer is obtained through applying a Gabor filter on the images.
- Max operation will be applied on the responses of S1 layer to produce cells with larger receptive fields in C1 layer.
- Within each filter band, a square of four adjacent, non-overlapping C1 units is then grouped to provide input to a S2 unit.
- The S2 units are again pooled by a max operation to yield C2 units.

In fact in this model [2], invariance to changes in the position of an optimal stimulus (within a range) is obtained by means of a maximum operation (max) performed on the simple cell inputs to the complex cells, where the strongest input determines the cell's output. In this model, using weighted sum more complex features has been made and using pooling mechanism, robust feature

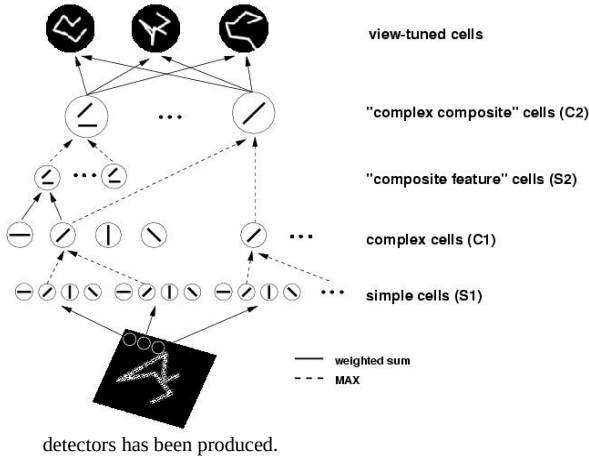


Figure 3 The standard HMAX model

### 3. The Proposed Model

In this part we introduce the proposed model to design a cooperative decision making method meeting two fundamental requirements i.e., selectivity and robustness. This model is a hierarchical model in which hierarchies are designed inspired by standard model of object recognition in primate cortex. Through the stages of the hierarchy, the knowledge of training data set will be extracted and similar knowledge will be grouped in some regions which are called clusters. These clusters in fact act like experts and do reasoning on any new data point.

Suppose we have a p-dimensional input data as below:

$$x = (x_1, x_2, \dots, x_p) \quad (1)$$

In the proposed model S1 layer (Fig. 3) is modeled by specifying a set of features for the data set. A set of features,  $F$  is defined for this data set as below:

$$F = (f_1, f_2, \dots, f_m) \quad (2)$$

where  $m$  is the number of assigned features to the data set. So in the first step we define  $m$  features for the data set (training data set). Using a characteristic function, a number in  $[0,1]$  is assigned to each data point for every feature. This number determines belonging degree of that data point to each feature. A larger value for every feature means higher degree of having that feature. In this model a Gaussian characteristic function has been used to specify the degree of having feature  $f_i$  for data point  $x$  :

$$G_{f_i}(x; \mu_{f_i}, \sigma_{f_i}) = \exp\left(-\frac{(x - \mu_{f_i})^2}{2\sigma_{f_i}^2}\right) \quad (3)$$

Like input data, a set of features is defined for output data and a characteristic function will be defined to specify the degree of having each feature for the output data.

$$G_{ff_i}(y; \mu, \sigma) = \exp\left(-\frac{(y - \mu_{ff_i})^2}{2\sigma_{ff_i}^2}\right)$$

where  $\mu_{f_i}$  and  $\sigma_{f_i}$  are the center and width of the  $i$ th feature of the input data and  $\mu_{ff_i}$  and  $\sigma_{ff_i}$  are the center and width of the  $i$ th feature of the output data. For a training data set which consists of an input and output, a set of features will be defined for both inputs and outputs. So in the first layer, for every data point we will have a set of numbers in  $[0, 1]$  that indicates the degree of having different features for input and the output of the mentioned data point. By having  $N$  training data points, the model produces a set of numbers in the first layer as below:

$$S_1: (g_{11}, g_{12}, \dots, g_{1m}), \dots, (g_{N1}, g_{N2}, \dots, g_{Nm}) \quad (4)$$

where  $g_{ij}$  is a number in  $[0,1]$  and indicates the degree of having  $j$ th feature for  $i$ th data point. So, in the first layer we have extracted different types of knowledge from training data set based on the features we have defined for the data set. Every part of knowledge like  $g_{ij}$  that has been characterized by a number in  $[0, 1]$  is called a cluster. Every cluster is characterized by an indicator knowledge that represents encapsulated data in that cluster. In the first layer, the indicator knowledge of every cluster is a number in  $[0, 1]$ , i.e.  $g_{ij}$ . In the further layers we try to make more composed knowledge about the data. In the second layer that is called clustering layer we try to find similarities between the extracted knowledge in the first layer and merge them to make more composed knowledge (features) of the data. By creating any new cluster the indicator knowledge must be updated. Suppose there are two clusters so-called  $C^1$  and  $C^2$  in the first layer that are represented as below:

$$C^1 == (g_{p1}) \text{ and } C^2 == (g_{q1}) \quad (5)$$

Where  $g_{p1}$  ( $i$ th feature of  $P$ th data point) is the indicator knowledge of  $C^1$  and  $g_{q1}$  ( $i$ th feature of  $Q$ th data point) is the indicator knowledge of  $C^2$ . Dissimilarity  $d(C^1, C^2)$  between  $C^1$  and  $C^2$  is computed from the set of pair wise knowledge indicators dissimilarities. In this model, we try to find the closest (least dissimilar) cluster considering the input knowledge and the output (labels) knowledge of the encapsulated data of the clusters:

$$d = \min_{\substack{i \in C^1 \\ j \in C^2}} d_{ij}$$

The algorithm of this approach has been demonstrated in Fig 5.

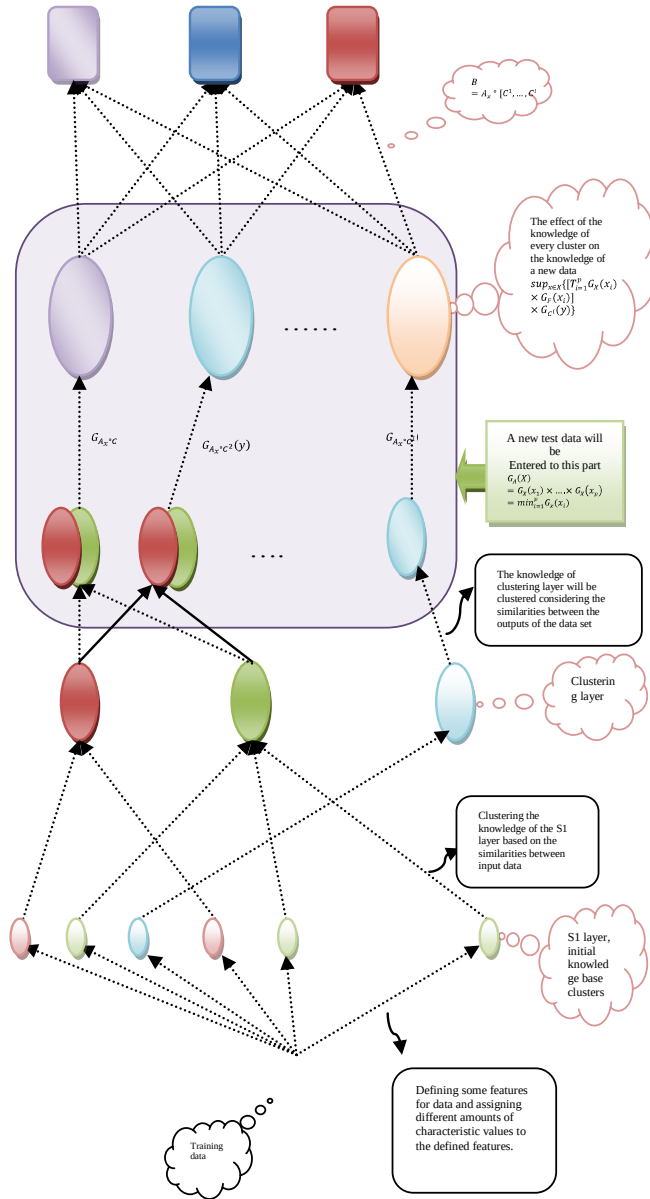


Figure 4 The structure of the proposed model

Applying this algorithm, we will make some clusters in second layer. The indicator knowledge is the maximum of the lower knowledge. So if two clusters  $C^1 = (g_{p_i})$  and  $C^2 = (g_{q_i})$  are merged, then the indicator knowledge of the new cluster is computed as below:

$$C^{1,2} = \max(g_{p_i}, g_{q_i}) \in [0,1] \quad (6)$$

In the next layer, another stage of clustering is done by finding the clusters that have the most similar features in the output of the data set. So here the features of the output of the training data will be compared and if two clusters are similar enough then they will be merged. The mentioned algorithm in figure 5 will be used again by this restriction that the comparison criterion is applied just on the

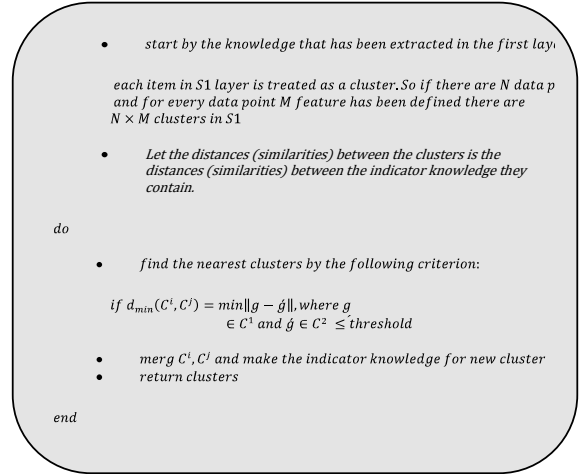


Figure 5 The algorithm of making composed knowledge in C1.

labels of the clusters (features of the outputs) that have been generated in the second layer. By using this algorithm, some clusters of the second layer will be merged to make some new clusters with new indicators in the third layer. To make a new indicator after merging two clusters, their indicator knowledge will be added to each other. The indicator of every cluster  $C^i$  in the third layer consists of the following knowledge:

$$C^i = (F(x_1), \dots, F(x_p), F(y_k)) \quad (7)$$

Where  $F(x_1)$  is the feature that has been defined for data point  $x_1$  and  $F(y_k)$  is the feature that has been defined for the output or the label of the training data. In the third layer, this knowledge has been shown as below:

$$G_{C^i}(x, y) = G((F_1^i \times F_2^i \times \dots \times F_p^i) \rightarrow C^i) = G_{(F_1^i \times F_2^i \times \dots \times F_p^i) \rightarrow C^i}(x, y) = G_{(F_1^i \times F_2^i \times \dots \times F_p^i)}(x) \times G_{C^i}(y) = G_{F_1^i}(x_1) \times \dots \times G_{F_p^i}(x_p) \times G_{C^i}(y) \quad (8)$$

$G$  is the characteristic function that has been defined for every feature for input and output data in the first layer. Through this process, we have clustered the data set to some knowledge based groups in a way that every cluster is able to transmit a part of knowledge about the training data considering the features and the outputs (labels) of the training data. Two last layers (fourth and fifth) layers are called inference layers and are responsible to reason any new data set to make decision regarding the output of the data point considering the experiences of the system. For every new input data set, feature extraction will be done in S1 layer as below:

$$G_A(X) = G_X(x_1) \times \dots \times G_X(x_p) \quad (9)$$

$G_X(x_i)$  is the characteristic function that has been defined for features that we have assigned to  $x_i$ . Every cluster  $C^i$  in third layer in fact is a set with partial knowledge about the behavior of the system. By inserting any new data point ( $G_A(X)$ ), every cluster (expert) tries to produce a proper output for the entered input considering its own knowledge. This process is done in fourth layer as below:

$$G_{B^i}(y) = G_{A_x \cdot C^i}(y) = \sup_{x \in X} [G_{A_x}(x) * G_{A^i \rightarrow C^i}(x, y)], \quad y \in Y \quad (12)$$

By substituting (8) and (9) in (10), we will have:

$$G_{B^i}(y) = G_{A_x \cdot C^i}(y) = \sup_{x \in X} [G_{A_x}(X) \times G_{A^i \rightarrow C^i}(X, y)] =$$

$$\sup_{x \in X} \{ [T_{i=1}^p G_X(x_i) \times [T_{i=1}^p G_F(x_i)] \times G_{C^i}(y) \} = \sup_{x \in X} \{ [T_{i=1}^p G_X(x_i) \times G_F(x_i)] \times G_{C^i}(y) \} \quad (13)$$

$G_{B^i}(y)$  is the response of  $i$ th cluster (expert), meeting the new data point.

If the number of clusters (experts) in the third layer is  $M$  then, in layer four there will be  $M$  responses. The final answer (decision) is made as below in the final layer.

$$B = A_x \circ [C^1, \dots, C^M] = \bigcup_{i=1}^M A_x \circ C^i \quad (14)$$

#### 4. Forecasting of Time-Series

Assume that  $s(k) (k = 1, 2, \dots, N)$  is a time series and  $x(k) = s(k) + n(k) (k = 1, 2, \dots, N)$  is the measured values of the time series and  $n(k)$  denotes noise. The goal of the forecasting of time-series is estimating of a future value of  $s$ , naming  $s(k+1)$ , having  $p$  past measurements of  $s(k)$ , which are  $x(k-p+1), x(k-p+2), \dots, x(k)$ . By having  $N$  data points,  $x_1, x_2, \dots, x(N)$ .  $D$  data points has been considered as training data set and  $N-D$  data points will be used as testing data set. The training data points are shown as below:

$$\begin{aligned} x^{(1)} &= [x(1), x(2), \dots, x(p)] \\ x^{(2)} &= [x(2), x(3), \dots, x(p+1)] \\ &\vdots \\ x^{(D-p)} &= [x(D-p), x(D-p+1), \dots, x(D)] \end{aligned}$$

The  $p$  first elements of  $x^{(1)}$  are the input data and the last element is the desired output. In this model we treat every data point as a feature and define the characteristic functions for these features by this assumption that every data is the mean of the characteristic function. So in a model with  $D$  training data points, there are  $D$  features and so we must define  $D$  characteristic function. As it was mentioned before, every data point will be the center of the characteristic function. So for every training data set which consists of  $p$  data points, there are  $D \times p$  characteristic value for the features of inputs and  $D$  characteristic values for the features of the output. Through the first layer, the training data set will be converted to a vector of values that are the characteristic values of different features for the data of that training set;

$$S_1: x^i \rightarrow ((g_{f_1}^{x_i}, g_{f_2}^{x_i}, \dots, g_{f_D}^{x_i}), (g_{f_1}^{x_{i+1}}, g_{f_2}^{x_{i+1}}, \dots, g_{f_D}^{x_{i+1}}), \dots, (g_{f_1}^{x_{i+p}}, g_{f_2}^{x_{i+p}}, \dots, g_{f_D}^{x_{i+p}}))$$

$g_{f_l}^{x_i}$  is the grade of having feature  $f_l$  where  $1 \leq l \leq D$  for the data point  $x_i$ . In layer  $S_1$  there will be  $D \times p$  clusters with the knowledge that was represented in (4). These features will be clustered again through the next layer, and will make some clusters with more composed knowledge. This clustering will be continued by considering the outputs of every cluster and will make the knowledge of the next layer. In the third layer, the knowledge we have learnt from training data has been stored as some clusters. We will use this knowledge to forecast the output of a new data.

In this work, we apply the model for a single-stage forecasting of a chaotic time-series obtained from the Mackey-Glass equations. From 1000 data points that were produced by Mackey-Glass equation, we took 500 to train the model and the other 500 to test it. The window size to make the training set is five and every training set is shown as below:

$$x^i = \{(x(i), x(i+1), x(i+2), x(i+3)), (x(i+4))\}$$

All the 500 data points in the training set were used to make 500 features and 500 characteristic functions. The characteristic function is a Gaussian function whose mean are the training data points. The standard deviation was initialized on 0.5 for all the characteristic functions. The evaluation was done based on root mean-square error:

$$RMSE_s = \sqrt{\frac{1}{500} \sum_{k=501}^{1000} [x(k+1) - f_x(x^k)]^2}$$

The results of the experiments with 500 training data and on 500 test data, with 500 characteristic functions has been shown in figure 6. In compare to other time series forecasters like neural networks the results of this method was better and faster. We plan to expand the work and do some experiments on noisy data to observe the robustness of the method.

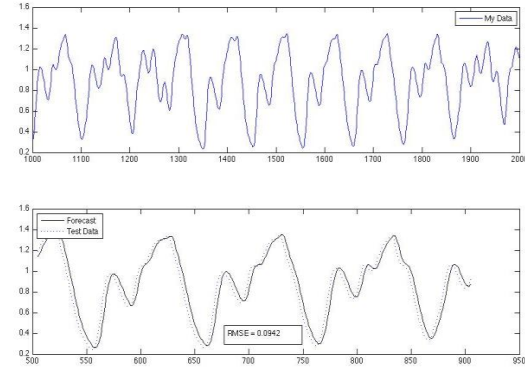


Fig 6. The experiments on 500 test data

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